



ROOT CAUSE UNLOCKED

The Complete Patient's Guide to *Knee* Pain

*A Whole-Body Approach to Understanding, Treating, and Recovering
from Knee Pain*

Dr. Adam Seay, DC

Root Cause Unlocked, Garner, NC · rootcauseunlocked.com · (919) 662-0520

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BEFORE YOU START

Disclaimer: *This ebook is for educational purposes only and does not constitute medical advice. Always consult with a qualified healthcare provider before starting any new treatment, supplement, exercise program, or dietary change. Do not start, stop, or change any prescribed medication without speaking to the prescriber. If you experience a hot, swollen joint with fever, sudden inability to bear weight after an injury, unexplained weight loss, or night pain that is not relieved by any position, seek medical attention promptly. This guide does not diagnose your knee pain. Only an examination by your provider can do that.*

A note on the evidence in this guide. *Every numbered citation points to a specific published study that was retrieved and checked against the source paper this session. Where the research is weak, mixed, or negative, this guide says so rather than overstating it. Where a claim reflects clinical experience rather than a settled body of research, that distinction is stated plainly in the text.*

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1. Understanding Knee Pain: Beyond “Wear and Tear”

If you have been living with knee pain, there is a good chance someone has already told you it is “just wear and tear,” and that the next step is to wait until it gets bad enough for a replacement. That message is common, and it is also incomplete. Knee pain is one of the most common reasons people see a doctor or a chiropractor, and osteoarthritis of the knee is a major driver of that burden worldwide. Global estimates put the number of people living with osteoarthritis at roughly 595 million in 2020, about 7.6 percent of everyone on earth, a figure that has more than doubled since 1990, and the knee is the single most commonly affected joint.¹ Those numbers are sobering, but they are also a reminder that you are far from alone, and that a great deal of research attention has gone into understanding what actually helps.

Here is the part that often gets left out of the “wear and tear” conversation. The knee is a hinge joint sitting in the middle of a long chain that runs from your foot to your pelvis and spine. It does not operate in isolation. What your hip is doing, what your ankle is doing, and even whether your legs are functionally the same length all influence how force moves through your knee with every step. A knee that hurts is not always a knee with a structural problem of its own. Sometimes it is an honest bystander, absorbing the consequences of dysfunction happening one or two joints away.

This guide is built around that idea. It will walk you through the anatomy of the knee and the common diagnoses associated with knee pain, and it will also walk you through the hip, the ankle, the pes anserine bursa on the inside of the knee, and the question of leg length, because a thorough evaluation of knee pain has to look beyond the knee itself. Throughout, this guide tries to tell you honestly what the research supports, what it does not, and where a claim reflects clinical judgment rather than a settled evidence base. You deserve accurate information, not reassurance that outruns the science, and not fear that outruns it either.

2. Anatomy of the Knee

Understanding the basic structure of your knee makes the rest of this guide easier to follow, and it also helps demystify a joint that can otherwise feel like a mystery box of pain.

The knee is really two joints working together inside one capsule. The **tibiofemoral joint** is the main hinge between your thighbone (femur) and shinbone (tibia). The **patellofemoral joint** is where your kneecap (patella) glides in a groove on the front of the femur as you bend and straighten the leg. Between the femur and tibia sit two C-shaped cartilage cushions, the **menisci**, which distribute load and add stability. The joint is held together by four major ligaments: the anterior and posterior cruciate ligaments (ACL and PCL) crossing inside the joint to control forward, backward, and rotational movement, and the medial and lateral collateral ligaments (MCL and LCL) on the inner and outer sides of the joint controlling side-to-side stress.

On the front of the thigh, the quadriceps muscle group narrows into the patellar tendon, which attaches below the kneecap and is the primary engine for straightening the knee. On the outer side of the thigh, the iliotibial (IT) band, a thick band of connective tissue, runs from the hip down past the outside of the knee. On the inner side, just below the joint line, three tendons, the sartorius, gracilis, and semitendinosus, converge onto the shinbone in a structure shaped like a goose's foot, which is why it is called the pes anserine, Latin for "goose foot." Underneath those tendons sits the pes anserine bursa, a small fluid-filled sac that reduces friction. You will hear much more about this structure in Chapter 8, because it is one of the most commonly overlooked sources of medial knee pain.

Finally, the knee does not act alone. It sits between the hip above and the ankle below, sharing muscles, fascia, and movement patterns with both. A change in how either of those joints moves changes the load the knee has to manage. Keep that image in mind. It is the thread running through much of this guide.

3. Common Causes of Knee Pain

Knee pain has many possible sources, and more than one can be present at the same time. Broadly, the common causes fall into a few categories.

Osteoarthritis is the gradual breakdown of the cartilage that cushions the tibiofemoral or patellofemoral joint, often accompanied by bone remodeling and mild inflammation. It is the cause most people have already heard about, and it is a real and common source of knee pain, particularly after middle age. It is also, as later chapters will show, frequently overdiagnosed as the explanation for pain that has a different, more treatable source.

Meniscus tears can happen from a twisting injury or can develop gradually as cartilage degenerates with age. Degenerative meniscus tears are extremely common on imaging in older adults, including in people with no knee pain at all, which is an important piece of context for Chapter 4.

Patellofemoral pain syndrome describes pain around or behind the kneecap, often worse with stairs, squatting, or prolonged sitting. It is frequently linked to how the kneecap tracks in its groove, which in turn is influenced by hip and, to a lesser degree, foot and ankle mechanics, the subject of Chapters 6 and 7.

Ligament sprains and tears, including ACL injuries, usually follow a specific traumatic event, often involving a pivot, a sudden stop, or a direct blow, and typically cause immediate swelling and instability.

Iliotibial band syndrome causes pain on the outer side of the knee, common in runners and cyclists, related to friction and tension in the IT band as it crosses the outside of the joint.

Bursitis is inflammation of one of the knee's fluid-filled cushioning sacs. Prepatellar bursitis (in front of the kneecap, sometimes called “housemaid’s knee”) is the most familiar version, but the pes anserine bursa on the inner side of the knee is a separate and commonly missed source of medial pain, covered in its own chapter here because generic osteoarthritis material tends to skip it entirely.

Inflammatory and metabolic causes, including gout, pseudogout, and inflammatory arthritis, can also produce knee pain and swelling and generally need their own medical workup rather than a self-directed approach.

The point of this list is not to hand you a diagnosis. It is to show you the range of what “knee pain” can mean, and to make clear that the source of your pain is a question worth answering carefully, ideally with hands-on examination, rather than assuming from age or an X-ray report alone.

4. How Knee Pain Is Diagnosed: The Bone-on-Bone Myth

If you have had an X-ray of an aching knee, there is a fair chance you left the appointment with a phrase that has stuck with you ever since: “bone on bone.” It is meant descriptively, referring to severe loss of the cartilage space between the femur and tibia on an X-ray, but for most patients it lands as a verdict, as though the knee is finished and nothing but a replacement will help. That reaction is understandable, and it is also not well supported by the evidence on how imaging relates to pain.

A 2025 study in the *Journal of the American Academy of Orthopaedic Surgeons* examined imaging and pain data from nearly 4,800 people and asked directly whether radiographic grading or detailed MRI scoring could predict a person’s pain and symptoms. The answer was no. Neither the standard radiographic grading system nor the detailed MRI scoring system used in the study reliably predicted who had pain and who did not, and the researchers found that structural imaging findings, even when carefully graded by established systems, explained only a small fraction of the variation in symptoms. The mismatch actually grew larger in people with more severe symptoms.³ In plain terms, two people can have nearly identical X-rays, and one can be in serious pain while the other is playing tennis. An X-ray showing severe joint space narrowing is a real finding, but it is not, by itself, a sentence, and it does not tell your provider or you exactly why your knee hurts today.

This matters even more once you understand the rest of this guide. If a person’s medial knee pain can come from pes anserine irritation, a hip that is not doing its share of the work, or an ankle that will not bend the way it needs to, then a frightening X-ray becomes even less determinative than most patients fear. A image can show real degenerative change and still be only part, or in some cases very little, of the actual story behind your pain.

For most knee pain without a specific injury or red flags (see Chapter 14), your provider’s clinical examination, your history, and how your knee moves and responds to specific tests carry more diagnostic weight day to day than imaging does. Imaging still has an important role, particularly when there has been trauma, when symptoms are not improving, or when surgery is being considered, but it is one piece of information, not the whole picture.

5. Regional Interdependence: Why the Knee Is Rarely the Whole Story

There is a formal name in rehabilitation science for the idea that the painful joint is not always the source joint. It is called **regional interdependence**, and it describes how impairments in one region of the body, sometimes a region that feels completely unrelated, can contribute to symptoms somewhere else entirely. A widely cited 2013 paper describing the model puts it this way: seemingly unrelated impairments in remote anatomical regions of the body may contribute to, and be associated with, a patient's primary complaint, and treating a distant region can produce a measurable effect on the area that actually hurts.²

This is the organizing idea behind how I evaluate knee pain, and it is why this guide devotes real space to the hip, the ankle, and leg length rather than treating the knee as a closed system. The knee is what mechanical engineers might call a “victim joint.” It has comparatively little rotational capacity of its own, which means when the joints above or below it move poorly, the knee often absorbs the resulting stress rather than the hip or ankle where the actual restriction lives. A hip that will not extend fully, or an ankle that will not bend forward past a certain point, changes the angle at which force arrives at the knee with every single step, thousands of times a day.

None of this means the knee itself is never the problem. It frequently is. Regional interdependence is not a reason to ignore the knee. It is a reason to also look above and below it, because treating a stiff hip or a restricted ankle can sometimes do more for a painful knee than anything aimed directly at the knee joint itself. The next three chapters take that idea and make it concrete: the hip, the ankle, and leg length, each in turn.

6. The Hip Connection

The hip and the knee share several muscles, most importantly the muscles that control the outside of the hip, the abductors and external rotators, along with the large muscle group on the back of the hip, the extensors (primarily the gluteus maximus). When those muscles are weak or slow to activate, the thigh tends to drift inward during standing, walking, and stair use, a pattern sometimes visible as the knee caving slightly toward the midline. That inward drift changes the angle of pull on the kneecap and increases stress on the inner structures of the joint.

The research connecting hip strength to knee pain is genuinely substantial, particularly for patellofemoral pain. A 2026 systematic review and meta-analysis in *Sports Medicine*, pooling 82 randomized trials and more than 4,000 participants with patellofemoral pain, examined whether gains in muscle strength over the course of treatment tracked with improvements in pain and function. They did, and the relationship was strongest for the hip. Increases in hip abductor strength showed the strongest association with pain reduction of any muscle group measured, followed closely by hip internal rotator and hip extensor strength, with knee extensor strength also showing a strong relationship.⁴ The authors concluded that the findings reinforce the clinical relevance of strength-focused rehabilitation, with particular attention to the hip.

For knee osteoarthritis specifically, the picture is still developing but points the same direction. A 2026 cross-sectional study of 112 adults compared people with and without symptomatic knee osteoarthritis and found that people with knee osteoarthritis had measurably weaker hip extensor strength than those without it, and that weaker hip extensors tracked with more pain, worse function, and more severe radiographic changes.⁵ Because the study was a single snapshot in time rather than a trial that followed people over years, it cannot prove that weak hips cause worse knee osteoarthritis rather than the reverse, and the authors say so directly, calling for longitudinal research to sort out the direction of that relationship. What it does establish is that the association is real and worth taking seriously in an evaluation, not an afterthought.

There is also direct evidence that training the hip changes outcomes at the knee. A 2026 randomized controlled trial gave collegiate athletes with patellofemoral pain a six week program built around hip and ankle mobility work, compared with athletes who continued their usual sport training alone. The group that received the hip and ankle program showed significantly greater improvement in pain and in knee-specific function scores, along with measurable gains in hip rotation range of motion and in a balance test that reflects lower-limb control.⁶ The trial was not large and the comparison group did not receive an alternative active treatment, both of which the authors acknowledge as limitations, but it is a genuine example of treating the hip (and ankle) producing a change at the knee, which is exactly what the regional interdependence model predicts.

What this means practically: if your knee pain, especially around or behind the kneecap, has not responded to treatment aimed only at the knee, hip strength and hip control are worth a real look. This is not a substitute for addressing the knee directly when the knee itself needs it. It is an additional place to look that generic knee-only advice tends to skip.

7. The Ankle Connection

The ankle's contribution to knee pain gets far less attention than the hip's, but the mechanical logic is the same. When you walk, run, squat, or descend stairs, your shin needs to travel forward over your foot, which requires a specific amount of ankle dorsiflexion, the motion of bringing your foot and shin closer together at the front of the ankle. When that range of motion is restricted, whether from old ankle sprains, tight calf muscles, or simple stiffness, the body still needs to get the shin forward, and it tends to compensate by pulling that motion from somewhere else in the chain, commonly the knee.

A systematic review published in the *Journal of Science and Medicine in Sport* looked specifically at what happens to lower-limb mechanics when ankle dorsiflexion is restricted during landing tasks. Reviewing six studies, the authors concluded that there is evidence restricted dorsiflexion range of motion can alter lower-extremity landing mechanics in ways that may predispose a person to injury, though they were candid that the studies disagreed on exactly which mechanical variables were most affected and called for more consistent research.⁷ That caveat is worth taking seriously. This is a real and biologically plausible connection, not a settled, precisely quantified one.

The evidence gets more mixed when the question narrows specifically to patellofemoral pain. A 2026 systematic review and meta-analysis pooling 42 studies and more than 2,300 participants compared foot and ankle biomechanics in people with and without patellofemoral pain. It found strong evidence for one specific foot pattern, greater “navicular drop,” a measure of how much the arch flattens under load, along with increased rearfoot motion during landing and step-down tasks in people with patellofemoral pain. Ankle dorsiflexion range of motion itself, however, did not differ significantly between the two groups in this particular pooled analysis.⁸ Read together with the hip and ankle mobility trial from the previous chapter, which did find that adding ankle mobility work to a rehab program improved outcomes, the honest summary is that the ankle's role in knee pain is real and worth assessing, but it likely runs through a combination of dorsiflexion restriction, foot arch behavior, and rearfoot motion rather than dorsiflexion range of motion in isolation. This is a case where the details of the mechanism are still being worked out even though the general direction, that ankle and foot mechanics influence the knee, is well supported.

What this means practically: a knee evaluation that never checks ankle mobility is skipping a plausible and testable contributor, particularly for anyone whose knee pain worsens with squatting, stairs, or lunging movements that demand a forward-traveling shin.

8. The Pes Anserine Bursa: A Commonly Missed Cause of Medial Knee Pain

This chapter exists because most general osteoarthritis material skips it, and skipping it does a disservice to a large group of people with pain on the inside of the knee. As Chapter 2 described, the pes anserine is where three tendons, the sartorius, gracilis, and semitendinosus, converge onto the shinbone just below and to the inside of the knee joint, a few centimeters below the joint line. Beneath those tendons sits a small bursa, a fluid-filled cushion, and when that bursa or the surrounding tendons become irritated, the result is pain in almost exactly the same location that people, and sometimes their providers, assume must be arthritis.

The clinical picture is fairly characteristic. Pain and tenderness sit on the inner side of the knee, a couple of finger-widths below the joint line rather than directly on it, and it is often provoked by climbing stairs, standing up from a seated position, or lying with the knees together at night. According to a clinical reference summary on the condition, it disproportionately affects middle-aged and overweight women, frequently coexists with osteoarthritis rather than replacing it as a diagnosis, and typically develops gradually rather than from a specific injury. The same source notes that the condition is generally self-limiting and tends to respond well to conservative measures such as activity modification, stretching, and targeted exercise.⁹

There is good reason to think this region deserves specific attention rather than being lumped into a general osteoarthritis diagnosis. A study using data from the Osteoarthritis Initiative, a large long-running cohort, examined whether tenderness on palpation over the anserine bursa region predicted meaningful outcomes over time. It did. People with tenderness in that area at the start of the study had more than double the odds of frequent knee pain, and people with tenderness in that area consistently across follow-up visits had more than triple the odds of frequent pain, along with a higher likelihood of eventually needing a knee replacement.¹⁰ The authors' practical takeaway was that clinicians should pay closer attention to this area during a physical examination, particularly in middle-aged and older women, because it may flag people at higher risk of progression who could benefit from earlier, more targeted intervention.

At the same time, honesty requires including a complicating finding. A 2022 study that compared clinical diagnoses of pes anserine tendinitis and bursitis against MRI as a reference standard found that clinical diagnosis alone performed poorly, correctly identifying true cases only a little over 40 percent of the time, with similarly limited ability to correctly rule the condition out.¹¹ That does not mean the condition is not real or not worth considering. It means that a tender spot on the inside of the knee is a genuinely useful clinical clue, exactly as the Osteoarthritis Initiative data suggests, but it should be treated as one piece of a fuller evaluation rather than a diagnosis made from that single finding alone, and it is a reasonable reason to ask your provider to specifically examine and, if needed, image that region before assuming a painful medial knee is necessarily arthritic joint-surface pain.

The bottom line: if your knee pain is on the inner side, a few centimeters below the joint line rather than directly on it, worse on stairs or when standing up, and your imaging shows only mild or moderate

degenerative change, or even none at all, the pes anserine region deserves a specific look before medial knee pain gets filed away as simply osteoarthritis.

9. Functional Leg Length Inequality

Leg length inequality is one of the more genuinely debated topics in musculoskeletal care, and it deserves to be presented that way rather than oversold in either direction.

Some degree of measurable leg length difference is close to universal. A widely cited two-part review of the literature found that anatomic (true, structural) leg length inequality is present in roughly 90 percent of people, with an average difference of about 5 millimeters, and concluded that small anatomic differences of this size do not appear to become clinically significant until the difference reaches somewhere around 20 millimeters, roughly three quarters of an inch.¹² The same author's companion paper on functional leg length, meaning an apparent difference that comes from how the pelvis, hips, or soft tissue are positioned rather than an actual difference in bone length, concluded that this functional asymmetry is a distinct phenomenon from true anatomic inequality, may relate to muscular tightness or imbalance rather than structure, and should generally be addressed before assuming a measured difference reflects true bony inequality.¹³ In other words, not every leg length difference found on a table is the same kind of difference, and the research specifically cautions against treating small measured differences as automatically meaningful.

Where the evidence becomes more directly relevant to knee pain is at larger, measured, structural differences. Using data from a large community-based cohort, researchers found that people with a limb length difference of 2 centimeters or more had significantly higher rates of knee osteoarthritis than people without such a difference, even after accounting for age, sex, weight, and prior joint problems.¹⁴ A follow-up study from the same research group, following people over roughly six years, found that a limb length difference of 2 centimeters or more was associated with a meaningfully higher risk of the knee osteoarthritis on the shorter side getting worse over time, though the same study did not find a statistically significant increase in hip osteoarthritis risk.¹⁵ Mechanistically, this makes sense: a study that used shoe inserts to simulate a mild leg length difference in people with existing knee osteoarthritis found that even a difference of under 1.5 centimeters measurably changed how force moved through the ankle, knee, hip, and pelvis during walking, altering loading patterns at the knee in ways that plausibly help explain why a shorter limb tends to develop worse arthritis over time.¹⁶

Here is the honest summary. For structural leg length differences at or above roughly 2 centimeters, there is real, published evidence tying the difference to knee osteoarthritis risk and progression, and a plausible mechanical explanation for why. For the smaller, more subtle functional differences that a hands-on clinical exam often identifies, meaning a difference driven by muscle tension or pelvic positioning rather than actual bone length, the research base is thinner and the clinical significance threshold from the available literature sits considerably higher than many of the differences typically found on examination. In my clinical experience, addressing smaller functional asymmetries, through targeted mobility and strengthening work rather than a lift or a structural fix, can still meaningfully change how a knee (or a hip, or a low back) feels and functions, even at magnitudes below where the anatomic literature draws its line for significance. That is a legitimate clinical observation worth discussing with your provider, and it is presented here as exactly that:

clinical judgment, not a settled research finding, and readers should understand the distinction between the two.

10. Supplements for Knee Health: What the Evidence Actually Says

This is the chapter where honesty matters most, because supplements cost money and it is tempting to overstate what they do. No supplement, injection, or therapy described anywhere in this guide regenerates cartilage. That claim shows up constantly in marketing, and it is not supported by the research discussed below. What follows is a straightforward look at what the trials actually found, including where the results are disappointing.

10.1 Glucosamine and Chondroitin

These are the most heavily marketed joint supplements in existence, and they deserve the most careful treatment. The definitive trial is the Glucosamine/chondroitin Arthritis Intervention Trial (GAIT), a large, National Institutes of Health funded, randomized, placebo- and drug-controlled trial of 1,583 people with knee osteoarthritis. Over 24 weeks, glucosamine alone, chondroitin sulfate alone, and the combination were all compared against both placebo and celecoxib (a prescription anti-inflammatory). The result for the overall group was clear: glucosamine and chondroitin sulfate, alone or combined, did not effectively reduce pain compared with placebo. An exploratory subgroup analysis suggested the combination might help people with moderate to severe pain specifically, but the authors were explicit that this was a secondary, hypothesis-generating finding, not the trial's primary result.¹⁷ A follow-up analysis from the same trial looked at whether two years of treatment slowed the actual structural loss of joint space on X-ray. It did not. No treatment group reached a meaningful difference from placebo in preventing joint space loss overall, though the authors noted a possible signal in the subgroup with milder starting radiographic changes.¹⁸

What this means for you: the largest, most rigorous trial of glucosamine and chondroitin for knee osteoarthritis found that, for most people, they do not meaningfully reduce pain and do not slow structural progression on X-ray. If you already take these and feel they help you personally, that is worth discussing with your provider rather than stopping abruptly, but this guide cannot honestly recommend starting them with an expectation of reliable benefit based on the best available trial evidence.

10.2 Turmeric and Curcumin

The evidence here is more encouraging, and it is evidence gathered specifically in knee osteoarthritis, which is unusual for a supplement chapter. A systematic review and meta-analysis of 11 randomized trials and just over 1,000 participants found that curcumin and boswellia formulations were both significantly better than placebo for pain relief and functional improvement in knee osteoarthritis, with curcumin performing comparably to NSAID pain relievers while causing significantly fewer gastrointestinal side effects. The authors were careful to add that the overall evidence base is still not large or high-quality enough to support a firm clinical recommendation, even though the direction of the findings is positive.¹⁹ A separate multicenter

trial of 367 people with knee osteoarthritis directly compared *Curcuma domestica* extract against ibuprofen over four weeks and found the extract to be non-inferior, meaning it performed about as well, with meaningfully fewer stomach-related side effects.²⁰

Dosage commonly used: 500 to 1,500 mg curcuminoids daily, ideally combined with piperine or in a more absorbable liposomal or phytosome form. **Caution:** may interact with blood thinners, and should be avoided in gallbladder disease. Discuss with your provider before starting.

10.3 Methylsulfonylmethane (MSM)

A small placebo-controlled pilot trial of 50 people with knee osteoarthritis found that MSM, taken as 3 grams twice daily for 12 weeks, produced significant improvements in pain and physical function compared with placebo, without serious side effects, though the authors themselves called it a pilot trial and said longer-term benefit and safety needed confirmation.²¹ A more recent, larger trial in 88 adults with mild knee pain, using a considerably lower dose of 2,000 mg daily over 12 weeks, also found a statistically significant improvement in knee-related quality of life scores compared with placebo.²²

Dosage used in trials: 2,000 to 6,000 mg daily. **Caution:** generally well tolerated in the trials reported here.

10.4 Vitamin D

Vitamin D is frequently promoted for joint health, and the most rigorous trial available does not support that promotion for knee osteoarthritis specifically. A two-year, placebo-controlled, randomized trial gave people with symptomatic knee osteoarthritis either a vitamin D supplement or placebo and tracked both pain and cartilage volume loss on MRI. Vitamin D levels rose substantially in the treatment group as expected, but knee pain improved similarly in both groups with no meaningful difference between them, and cartilage volume loss was essentially the same in both groups as well.²³

What this means for you: correcting a genuine vitamin D deficiency remains worthwhile for bone health and overall wellness, and it is reasonable to have your level checked, but this trial found no evidence that supplementing vitamin D slows knee osteoarthritis or reduces its pain.

10.5 Omega-3 Fish Oil

Omega-3 fatty acids are commonly recommended as a general anti-inflammatory supplement, and the specific knee osteoarthritis trial evidence is more nuanced than a simple “more is better” story. A two-year randomized trial directly compared a high dose of fish oil (4.5 grams of omega-3 daily) against a low dose (0.45 grams daily) in 202 people with knee osteoarthritis. By two years, the low-dose group actually showed somewhat greater improvement in pain and function scores than the high-dose group, and the researchers concluded there was no additional benefit from the higher dose. They noted the low-dose formulation’s unexpectedly strong performance deserves further investigation rather than being taken as proof that less is better.²⁴

What this means for you: omega-3 fish oil is generally safe and has a reasonable overall health rationale, but the trial evidence specific to knee osteoarthritis does not support the idea that a higher dose produces a better outcome, and it should not be marketed or taken with an expectation of dramatic pain relief.

The honest summary of this chapter

Of the five supplements reviewed here, the strongest and most direct trial evidence, glucosamine and chondroitin, is negative for both pain and structural progression. Vitamin D's dedicated knee trial is also negative. Curcumin and MSM both have real, positive, knee-specific trial evidence, though the authors of that research describe it as promising rather than definitive. Fish oil is reasonable for general health but does not show a clear dose-dependent benefit for the knee specifically. None of this evidence, positive or negative, involves cartilage regrowth, and no claim to that effect appears anywhere in this guide.

11. Movement, Exercise, and Manual Care for the Knee

The single most consistently supported intervention in this entire guide is exercise. That is not a hedge or a consolation prize. It is the one treatment every major guideline body recommends for knee osteoarthritis, and no other option in this guide has been tested anywhere near as often.

A Cochrane systematic review, one of the most rigorous forms of evidence synthesis available, has asked this question repeatedly, and its current edition was published in December 2024, pooling 139 randomized trials and 12,468 participants. That update is more cautious than the earlier one it replaced, and the caution is worth understanding rather than glossing. Compared with a placebo or attention control, exercise likely improves physical function on moderate-certainty evidence and may reduce pain on low-certainty evidence; compared with usual care, the improvements run somewhat larger. The reviewers then compared their own results against the smallest change a patient would actually notice, which on a 0 to 100 scale is 12 points for pain and 13 for function, and found their confidence intervals either fell short of those thresholds or spanned both sides of them.²⁵ The fair reading is that exercise helps and that its average effect sits near the edge of perceptibility, which is a reason to do it properly rather than a reason to dismiss it. Both major guideline bodies for osteoarthritis treatment agree. The Osteoarthritis Research Society International (OARSI) lists structured, land-based exercise as a core treatment for knee osteoarthritis, meaning it belongs in essentially every treatment plan regardless of other factors.²⁶ The American College of Rheumatology and Arthritis Foundation's joint guideline gives exercise its strongest possible recommendation category for knee and hip osteoarthritis.²⁷

11.1 What kind of exercise

The research does not point to one single superior exercise style. What matters most is consistency and a program that addresses the whole chain discussed in this guide, not the knee in isolation.

Quadriceps strengthening remains foundational, since the quadriceps is the primary muscular support for the front of the knee and the patellofemoral joint. Straight leg raises, wall sits, and controlled mini-squats are reasonable starting points, progressing in range and resistance as tolerated.

Hip strengthening, given the evidence in Chapter 6, deserves equal billing with knee-focused work rather than being an afterthought. Side-lying leg lifts, clamshells, and glute bridges target the hip abductors and extensors most closely linked to knee pain outcomes in the research.

Ankle mobility work, given Chapter 7, is a reasonable addition for anyone whose knee pain worsens with squatting, lunging, or stairs. A simple wall-facing ankle mobility drill, tracking the knee toward the wall while keeping the heel down, performed for a few minutes daily, is a low-risk way to work on dorsiflexion range.

Low-impact aerobic activity, including walking, cycling, and water-based exercise, supports overall joint health, weight management (Chapter 13), and cardiovascular fitness without the repetitive impact that can aggravate an irritated joint.

11.2 Manual and hands-on care

Hands-on care, including chiropractic adjustment, joint mobilization, and soft tissue work, fits naturally within the regional interdependence framework described in Chapter 5. Because hip and ankle restrictions can drive knee symptoms, restoring motion at those joints through manual techniques, alongside an active exercise program rather than as a substitute for one, is a reasonable and commonly used part of a comprehensive plan. The strongest evidence in this guide is for active exercise; manual therapy is best understood as a way to help a stiff or restricted joint move well enough to get the most out of that exercise, rather than as a stand-alone fix.

Realistic expectation: exercise is not a quick fix, and most people need several weeks of consistent effort before noticing a clear change. It is, however, the intervention in this entire guide with the most consistent and highest-quality supporting evidence, and it is available to nearly everyone regardless of what is ultimately found to be driving their pain.

12. Surgery: What the Evidence Actually Shows

Surgery is sometimes the right answer for knee pain, and sometimes it is recommended for situations where the evidence does not actually support it. Both of those things are true, and this chapter tries to separate them honestly.

12.1 Arthroscopic surgery for degenerative knee disease

This is one of the most important corrections a patient can hear, because arthroscopic “clean-up” surgery, sometimes called débridement or lavage, has been recommended to enormous numbers of patients with degenerative knee pain over the decades, often on the logical-sounding premise that removing damaged tissue and flushing out debris should help. The landmark trial testing that premise found otherwise. In a randomized, placebo-controlled trial published in the *New England Journal of Medicine*, 180 patients with osteoarthritis of the knee were randomly assigned to arthroscopic débridement, arthroscopic lavage, or a placebo surgery involving skin incisions with no actual arthroscope inserted. Patients and the people assessing their outcomes did not know which group they were in. Over 24 months, neither real procedure produced better pain or function outcomes than the placebo surgery.²⁹ A second, independent randomized trial in 178 patients with moderate to severe knee osteoarthritis, comparing arthroscopic surgery plus optimized physical and medical therapy against optimized physical and medical therapy alone, found the same thing: no additional benefit from the surgery at two years on standard pain and function measures.³⁰

What this means for you: for degenerative knee disease without a specific mechanical problem like a locked or displaced meniscus fragment, arthroscopic clean-up surgery has been tested twice, rigorously, against doing nothing surgical, and it did not win. If arthroscopic surgery is recommended to you for arthritis-related knee pain, it is entirely reasonable to ask your surgeon how your specific situation differs from the patients in these two trials.

12.2 Total and partial knee replacement

Joint replacement is a different procedure addressing a different problem, and it should not be judged by the arthroscopy trials above. The evidence base here is thinner than many people assume, largely because it is difficult to run a placebo-controlled trial of a major joint replacement. A 2026 Cochrane review searching specifically for randomized trials comparing total or partial knee replacement against non-surgical care for moderate to severe osteoarthritis found only one qualifying trial, a Danish study of 100 participants comparing knee replacement plus a non-surgical program against the non-surgical program alone. In that trial, surgery produced a clinically meaningful reduction in pain at one year, with probably no meaningful difference in overall quality of life, and a higher rate of serious adverse events in the surgical group. The review’s authors rated the certainty of this evidence as very low, given that it rests on a single trial, and urged that the findings be interpreted cautiously.³¹

What this means for you: knee replacement can provide real pain relief for the right candidate with advanced disease, and it remains an appropriate option after conservative care has been genuinely tried, but the formal evidence base comparing it head to head against non-surgical treatment is smaller and less certain than most people assume. It is a major, largely irreversible procedure, and a decision worth making after the approaches in Chapters 6 through 11 have had a real trial, not before.

13. Diet, Weight, and Lifestyle

What you eat and how much load your body carries both influence your knee, and the evidence for weight is considerably stronger and more direct than the evidence for any specific food or diet pattern.

13.1 Weight and Load

Every pound of body weight translates into several pounds of additional force across the knee with each step, which makes weight a mechanical issue for this joint in a very literal sense, not just a general health one. The best evidence for a combined approach comes from the IDEA trial, a randomized study of 454 overweight and obese older adults with knee osteoarthritis, comparing a structured diet-and-exercise program, diet alone, and exercise alone over 18 months. The combined diet-and-exercise group lost significantly more weight than either single-intervention group, had lower knee joint loads during walking, had lower blood markers of inflammation, and reported better pain and function scores.²⁸ This is one of the more directly persuasive pieces of evidence in this entire guide, precisely because it measured mechanical joint load rather than relying on symptom reports alone. Both major osteoarthritis guidelines reflect this: OARSI lists dietary weight management alongside exercise as a core treatment for knee osteoarthritis specifically, and the American College of Rheumatology gives a strong recommendation to weight loss for people with knee or hip osteoarthritis who are overweight or obese.^{26,27}

What this means for you: modest, sustainable weight loss combined with exercise, rather than either alone, has some of the best and most mechanically direct evidence in this entire guide.

13.2 An Anti-Inflammatory Eating Pattern

There is no single food that fixes a knee, and this guide will not pretend otherwise. A generally anti-inflammatory pattern of eating, similar to what is often called a Mediterranean-style diet, supports overall metabolic and joint health as part of the broader picture above. That means emphasizing colorful vegetables and fruits, fatty fish and other quality protein sources, olive oil, nuts, and whole foods, while minimizing highly processed foods, refined sugar, and excessive alcohol. Think of this as sensible, general nutrition that supports the weight management and reduced systemic inflammation discussed above, not as a proven, stand-alone treatment for knee pain on its own.

13.3 Sleep and Stress

Poor sleep and high stress both amplify how pain is perceived and processed by the nervous system, and both make it harder to stay consistent with the exercise program that Chapter 11 identifies as the single best-supported intervention available. Prioritizing a consistent sleep schedule, limiting screens before bed, and building in some form of regular stress management, whether that is walking, breathing exercises, or simply

time away from constant demands, is not a treatment for knee pain specifically, but it removes obstacles that make every other part of this guide harder to follow through on.

14. The In-Clinic Pre-Treatment Protocol

Understand what this is for, because the distinction matters.

None of this rebuilds tissue. The loading and strengthening work in the previous chapters does that. What this protocol does is reduce the muscle tone, stiffness and tissue sensitivity that make the strength work harder to tolerate and less pleasant to do, particularly in the early weeks when things are at their most irritable.

It is preparation, not treatment. I use it in the clinic before the working part of a session, and I give it to patients to use at home before theirs. Patients who do it tend to tolerate the loading better and stick with it longer, and sticking with it is the actual mechanism by which recovery happens.

If you have to choose between doing this and doing your strength work, do your strength work.

Foam rolling.

- Quadriceps, prone, from hip to above the knee, 90 to 120 seconds per leg, angling the leg in and out.
- Hamstrings, seated, from the glute fold to above the knee, 90 to 120 seconds per leg, turning the leg in and out.
- Adductors, prone with the leg bent to 90 degrees, from groin to above the knee, 60 to 90 seconds per leg. Start light.

Percussion device (Hypervolt).

- Quadriceps, round ball or dampener, speed 2 to 3, circles, 60 to 90 seconds per leg. Avoid the kneecap and the joint line.
- IT band, proximal and distal, dampener or fork, level 3, linear strokes, 45 to 60 seconds per leg. Avoid the bony outside of the knee.
- Patellar tendon region, dampener, level 1, very gentle circles around the kneecap, 30 to 45 seconds per leg. Minimal pressure, no direct tendon contact.
- Pes anserine area, dampener, level 1 to 2, gentle, medial knee below the joint line, 30 to 45 seconds per leg.

The Stick.

- Quadriceps, hip to knee, 60 to 90 seconds per leg, separate passes for each muscle.
- Hamstrings, buttock to knee, 60 to 90 seconds per leg. Avoid the knee crease.
- IT band, standing, firm, hip to knee, 45 to 60 seconds per leg.

Additional cautions for the knee. Never apply direct pressure to the kneecap or the joint line. Avoid the hollow behind the knee entirely.

Safety, and I mean these.

- Avoid all of this over an acutely injured area in the first 48 to 72 hours.
- Reduce pressure over anything actively inflamed.
- Skip percussion therapy entirely if you have a pacemaker or another implanted electronic device.
- Avoid aggressive pressure over varicose veins or compromised skin.
- Start light. Pain must not exceed 6 out of 10. Back off immediately for sharp or shooting pain.
- Stop and seek advice if you get numbness, tingling, or weakness.
- If you are on a blood thinner, use lighter pressure throughout and check with your prescriber first.

One absolute rule before you roll or percuss a calf. Never work on a calf that is swollen, warm, or tight in one leg only, particularly after travel, surgery, or a period of immobilisation. That is a blood clot concern. It needs assessing the same day and it must never be massaged.

15. Red Flags: When to Seek Immediate Care

Most knee pain is not an emergency, and most of what is discussed in this guide can be worked through with your provider over a reasonable timeline. A small number of situations need prompt or emergency attention.

Seek emergency care if you experience:

- A hot, swollen, red knee accompanied by fever. This can indicate a septic (infected) joint, which is a genuine medical emergency requiring immediate evaluation.
- Sudden inability to bear any weight on the leg after a fall, twist, or other injury.
- A visibly deformed joint after trauma.
- Signs of a blood clot, such as calf swelling, warmth, and redness, particularly after recent surgery, injury, or prolonged immobility.

Contact your provider promptly if you experience:

- Unexplained weight loss along with knee pain.
- Night pain that is not relieved by any position or rest, and does not improve with usual measures.
- A knee that locks, gives way repeatedly, or will not fully straighten.
- Significant new swelling without a clear cause.
- Pain that is steadily worsening over several weeks despite rest and conservative care.

None of the information in this guide is meant to delay appropriate medical evaluation for any of the situations above.

16. Building Your Personal Recovery Plan

Use this as a starting framework to discuss with your provider, not as a substitute for an individualized evaluation. Every knee, and every chain leading up to and down from it, is a little different.

Getting an accurate picture

Ask specifically whether your hip strength, hip control, ankle mobility, and leg length have been assessed as part of your evaluation, alongside a direct examination of the knee itself, including the pes anserine region if your pain sits on the inner side of the joint. If your knee pain has been explained entirely by an X-ray without a hands-on exam of these other areas, it is entirely reasonable to ask for a more complete look before assuming the story is finished.

Weeks 1 to 3: Calm and Assess

- Begin gentle, low-impact movement such as walking or stationary cycling as tolerated.
- Start a basic hip and ankle mobility routine alongside gentle quadriceps activation.
- Discuss with your provider whether pes anserine involvement, hip weakness, ankle restriction, or leg length asymmetry appear to be contributing, rather than assuming a single explanation.
- Begin an anti-inflammatory eating pattern and prioritize sleep.

Weeks 4 to 8: Build

- Progress hip and quadriceps strengthening in frequency and resistance.
- Continue ankle mobility work, particularly if stairs, squatting, or lunging remain aggravating.
- Add structured, low-impact aerobic activity several times weekly.
- If weight is a relevant factor, begin working toward gradual, sustainable loss alongside the exercise program, not as a separate later project.

Weeks 9 and beyond: Sustain

- Maintain hip, ankle, and quadriceps work as an ongoing habit, not a temporary fix.
- Reassess with your provider if progress has stalled, since a stalled recovery is itself useful information about what might still be missing from the picture.
- Keep exercise, weight management, and manual care (if it has been helping) as the core, evidence-backed foundation of your plan, and treat supplements, if used at all, as a minor addition rather than the centerpiece.

Final Thoughts

Knee pain is common, and it is rarely as simple as the phrase “wear and tear” makes it sound. Your knee sits in the middle of a chain that includes your hip, your ankle, and the overall alignment of your legs, and a thorough evaluation looks at that whole chain rather than stopping at the joint that happens to hurt. The pes anserine bursa, a structure most general osteoarthritis material never mentions, is a genuinely common and often treatable source of medial knee pain that deserves its own consideration. A frightening X-ray is real information, but it is not, on its own, a reliable predictor of your pain or your future.

The best-supported tools available, exercise addressing the hip and ankle as well as the knee, gradual weight management where relevant, and an honest look at the structures most often overlooked, are available to nearly everyone, regardless of what imaging shows. Where the evidence for a supplement or a procedure is weak or negative, this guide has tried to say so plainly, because an accurate picture, even an imperfect one, serves you better than a reassuring one that overstates what the research actually supports.

If you would like an evaluation that looks at your knee as part of a whole functional chain, hip, ankle, leg length, and the knee itself, I offer a free Root Discovery Session at Root Health to talk through what may be contributing to your pain and what a reasonable next step looks like. You can schedule one at rootcauseunlocked.com/discovery. You can also reach the office at (919) 662-0520.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

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This guide was compiled from peer-reviewed medical literature and clinical experience. Every citation was retrieved and verified against the published record this session. It is intended to give you knowledge and options to discuss with your healthcare provider. It is not a substitute for individual medical care, and it does not promise a cure.

This is a DRAFT pending Dr. Seay's clinical review.

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